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22CS907

CLOUD FOUNDATIONS

Department : CSE

Batch/Year : 2023-2027 / II Year

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2. COURSE OBJECTIVES

- To describe the different ways a user can interact with Google Cloud.
- To discover the different compute options in Google Cloud and implement a variety of structured and unstructured storage models.
- To confer the different application managed service options in the cloud and outline how security in the cloud is administered in Google Cloud.
- To demonstrate how to build secure networks in the cloud and identify cloud automation and management tools.
- To determine a variety of managed big data services in the cloud.

3. PRE REQUISITES

- **Pre-requisite Chart**

22CS907 – CLOUD FOUNDATIONS



**22CS203– DATABASE
MANAGEMENT SYSTEMS**

4. SYLLABUS

22CS907

CLOUD FOUNDATIONS

L	T	P	C
2	0	2	3

UNIT I INTRODUCTION TO CLOUD

6 + 6

Cloud Computing - Cloud Versus Traditional Architecture - IaaS, PaaS, and SaaS - Cloud Architecture - The GCP Console - Understanding projects - Billing in GCP - Install and configure Cloud SDK - Use Cloud Shell - APIs - Cloud Console Mobile App.

List of Exercise/Experiments:

1. Install and configure cloud SDK.
2. Connect to computing resources hosted on Cloud via Cloud Shell.

UNIT II COMPUTE AND STORAGE

6 + 6

Compute options in the cloud - Exploring IaaS with Compute Engine - Configuring elastic apps with autoscaling - Exploring PaaS - Event driven programs - Containerizing and orchestrating apps - Storage options in the cloud - Structured and unstructured storage in the cloud - Unstructured storage using Cloud Storage - SQL managed services - NoSQL managed services.

List of Exercise/Experiments:

1. Create virtual machine instances of various machine types using the Cloud Console and the command line. Connect an NGINX web server to your virtual machine.
2. Create a small App Engine application that displays a short message.
3. Create, deploy, and test a cloud function using the Cloud Shell command line.
4. Deploy a containerized application.
5. Create a storage bucket, upload objects to it, create folders and subfolders in it, and make objects publicly accessible using the Cloud command line.

UNIT III APIs AND SECURITY IN THE CLOUD

6 + 6

The purpose of APIs – API Services - Managed message services - Introduction to security in the cloud - The shared security model - Encryption options - Authentication and authorization with Cloud IAM - Identify Best Practices for Authorization using Cloud IAM.

List of Exercise/Experiments:

1. Deploy a sample API with any of the API service.
2. Publish messages with managed message service using the Python client library.
3. Create two users. Assign a role to a second user and remove assigned roles associated with Cloud IAM. Explore how granting and revoking permissions works from Cloud Project Owner and Viewer roles.

UNIT IV NETWORKING, AUTOMATION AND MANGAEMENT TOOLS 6 + 6

Introduction to networking in the cloud - Defining a Virtual Private Cloud - Public and private IP address basics - Cloud network architecture - Routes and firewall rules in the cloud - Multiple VPC networks - Building hybrid clouds using VPNs - Different options for load balancing - Introduction to Infrastructure as Code - Terraform - Monitoring and management tools.

List of Exercise/Experiments:

1. Create several VPC networks and VM instances and test connectivity across networks.
2. Create two nginx web servers and control external HTTP access to the web servers using tagged firewall rules.
3. Configure a HTTP Load Balancer with global backends. Stress test the Load Balancer and denylist the stress test IP.
4. Create two managed instance groups in the same region. Then, configure and test an Internal Load Balancer with the instances groups as the backends.
5. Monitor a Compute Engine virtual machine (VM) instance with Cloud Monitoring by creating uptime check, alerting policy, dashboard and chart.

UNIT V BIG DATA AND MACHINE LEARNING SERVICES 6 + 6

Introduction to big data managed services in the cloud - Leverage big data operations - Build Extract, Transform, and Load pipelines - Enterprise Data Warehouse Services - Introduction to machine learning in the cloud - Building bespoke machine learning models with AI Platform - Pre-trained machine learning APIs.

List of Exercise/Experiments:

1. Create a cluster, run a simple Apache Spark job in the cluster, then modify the number of workers in the cluster.
2. Create a streaming pipeline using one of the cloud service.
3. Set up your Python development environment, get the relevant SDK for Python, and run an example pipeline using the Cloud Console.
4. Use cloud-based data preparation tool to manipulate a dataset. Import datasets, correct mismatched data, transform data, and join data.
5. Utilize a cloud-based data processing and analysis tool for data exploration and use a machine learning platform to train and deploy a custom TensorFlow Regressor model for predicting customer lifetime value.

TOTAL: 60 PERIODS

5. COURSE OUTCOME

At the end of this course, the students will be able to:

CO1: Describe the different ways a user can interact with Google Cloud.

CO2: Discover the different compute options in Google Cloud and implement a variety of structured and unstructured storage models.

CO3: Discuss the different application managed service options in the cloud and outline how security in the cloud is administered in Google Cloud.

CO4: Demonstrate how to build secure networks in the cloud and identify cloud automation and management tools.

CO5: Discover a variety of managed big data services in the cloud.

6. CO - PO / PSO MAPPING

CO	HKL	PROGRAM OUTCOMES												PSO		
		K3	K4	K5	K5	K3,K4,K5	A3	A2	A3	A3	A3	A3	A2	PSO1	PSO2	PSO3
		PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12			
C306.1	K3	2	1	1	-	-	-	-	-	-	2	2	2	2	2	2
C306.2	K3	3	3	3	-	-	-	-	2	2	2	2	2	2	2	2
C306.3	K3	3	3	3	-	-	2	-	2	2	2	2	2	2	2	2
C306.4	K3	3	3	3	-	-	-	-	2	2	2	2	2	2	2	2
C306.5	K3	3	3	3	-	-	2	-	-	2	2	2	2	2	2	2

Correlation Level:

1. Slight (Low)

2. Moderate (Medium)

3. Substantial (High)

If there is no correlation, put "-".

7. LECTURE PLAN

Sl. No.	Topic	Number of Periods	Proposed Date	Actual Lecture Date	CO	Taxonomy Level	Mode of Delivery
1	Introduction to big data managed services in the cloud	1	19.03.2025		C05	K2	PPT
2	Leverage big data operations	1	20.03.2025		C05	K3	PPT
3	Build Extract, Transform, and Load pipelines	1	21.03.2025		C05	K3	PPT
4	Enterprise Data Warehouse Services	1	22 .03.2025		C05	K3	PPT
5	Introduction to machine learning in the cloud	1	24.03.2025		C05	K2	PPT
6	Building bespoke machine learning models with AI Platform, Google's pre-trained machine learning APIs	1	26.03.2025		C05	K3	PPT

8. ACTIVITY BASED LEARNING

1.Design and deploy a custom ML model on AI platforms using Google Vertex AI?

2.Design a cloud-based big data architecture for a real-world application (e.g., fraud detection, recommendation system)

3. Design a basic data warehouse schema for an e-commerce company. Identify key fact and dimension tables.?

4. Act as a data architect and present a data warehouse modernization plan to a company's leadership team.

5. Suppose a retail company wants to build a demand forecasting model using AI Platform. Outline the steps needed to prepare the data, train the model, and deploy it.

9. UNIT V - LECTURE NOTES

BIG DATA AND MACHINE LEARNING SERVICES

INTRODUCTION TO BIG DATA MANAGED SERVICES IN THE CLOUD

What is Big Data?

Big Data is a collection of datasets that is large and complex, making it very difficult to process using legacy data processing applications.

So, legacy or traditional systems cannot process a large amount of data in one go.

Tools for big data can help with the volume of the data collected, the speed at which that data becomes available to an organization for analysis, and the complexity or varieties of that data.

Data can be a company's most valuable asset. Using big data to reveal insights can help understand the areas that affect your business—from market conditions and customer purchasing behaviors to your business processes.

Big Data Overview:

Big data consists of petabytes (more than 1 million gigabytes) and exabytes (more than 1 billion gigabytes), as opposed to the gigabytes common for personal devices.

As big data emerged, so did computing models with the ability to store and manage it. Centralized or distributed computing systems provide access to big data. Centralized computing means the data is stored on a central computer and processed by computing platforms like BigQuery.

Distributed computing means big data is stored and processed on different computers, which communicate over a network. A software framework like Hadoop makes it possible to store the data and run applications to process it.

There are benefits to using centralized computing and analyzing big data where it lives, rather than extracting it for analysis from a distributed system. Insights are accessible to every user in your company—and integrated into daily workflows—when big data is housed in one place and analyzed by one platform.

Characteristics of big data:

Big data is different from typical data assets because of its volume complexity and need for advanced business intelligence tools to process and analyze it. The attributes that define big data are volume, variety, velocity, and variability. These big data attributes are commonly referred to as the four v's.

Volume: The key characteristic of big data is its scale—the volume of data that is available for collection by your enterprise from a variety of devices and sources.

Variety: Variety refers to the formats that data comes in, such as email messages, audio files, videos, sensor data, and more. Classifications of big data variety include structured, semi-structured, and unstructured data.

Velocity: Big data velocity refers to the speed at which large datasets are acquired, processed, and accessed.

Variability: Big data variability means the meaning of the data constantly changes. Therefore, before big data can be analyzed, the context and meaning of the datasets must be properly understood.

Types of Big Data:

Big Data is essentially classified into three types:

- ✓ Structured Data
- ✓ Unstructured Data
- ✓ Semi-structured Data

The above three types of Big Data are technically applicable at all levels of analytics. It is critical to understand the source of raw data and its treatment before analysis while working with large volumes of big data. Because there is so much data, extraction of information needs to be done efficiently to get the most out of the data.

OVERVIEW OF BIG DATA MANAGED SERVICES:

- ✓ Dataproc
- ✓ Dataflow
- ✓ BigQuery

Google Cloud Dataproc

Cloud Dataproc is a managed Apache Hadoop and Spark service for batch processing, querying, streaming and machine learning. Users can quickly spin up Hadoop or Spark clusters and resize them at any time without compromising data pipelines through automation and orchestration. It can be fully integrated with other Google big data services, such as BigQuery and Bigtable, as well as Stackdriver Logging and Monitoring.

Google Cloud Dataflow:

Cloud Dataflow is a serverless stream and batch processing service. Users can build a pipeline to manage and analyze data in the cloud, while Cloud Dataflow automatically manages the resources. It was built to integrate with other Google services, including BigQuery and Cloud Machine Learning, as well as third-party products, such as Apache Spark and Apache Beam.

Google BigQuery:

BigQuery is a data warehouse that processes and analyzes large data sets using SQL queries. These services can capture and examine streaming data for real-time analytics. It stores data with Google's Capacitor columnar data format, and users can load data via streaming or batch loads. To load, export, query and copy data, use the classic web UI,

the web UI in the GCP Console, the bq command-line tool or client libraries. Since BigQuery is a serverless offering, enterprises only pay for the storage and compute they consume.

LEVERAGE BIG DATA OPERATIONS WITH CLOUD DATAPROC

DATAPROC:

Dataproc is a managed Spark and Hadoop service that lets you take advantage of open source data tools for batch processing, querying, streaming, and machine learning.

Dataproc is a fully managed service for hosting open source distributed processing platforms such as Apache Spark, Presto, Apache Flink and [Apache Hadoop](#) on Google Cloud. Unlike on-premise clusters, Dataproc provides organizations the flexibility to provision and configure clusters of varying size on demand.

Dataproc automation helps to create clusters quickly, manage them easily, and save money by turning clusters off when you don't need them.

In addition, Dataproc has powerful features to enable your organization to lower costs, increase performance and streamline operational management of workloads running on the cloud.

Hadoop and Spark:

Apache Hadoop and Apache Spark are open source technologies that often are the foundation of big data processing.

Apache Hadoop is a set of tools and technologies which enables a cluster of computers to store and process large volumes of data. The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models.

Apache Spark is a unified analytics engine for large-scale data processing and achieves high performance for both batch and stream data.

Cloud Dataproc features:

Cost effective: Dataproc is priced at 1 cent per virtual CPU per cluster per hour, on top of any other Google Cloud resources you use. In addition, Dataproc clusters can include preemptible instances that have lower compute prices. You use and pay for things only when you need them.

Fast and scalable: Dataproc clusters are quick to start, scale, and shut down, and each of these operations takes 90 seconds or less, on average. Clusters can be created and scaled quickly with many virtual machine types, disk sizes, number of nodes, and networking options.

Open source ecosystem: You can use Spark and Hadoop tools, libraries, and documentation with Dataproc.

Fully managed: You can easily interact with clusters and Spark or Hadoop jobs, without the assistance of an administrator or special software, through the Cloud Console, the Google Cloud SDK, or the Dataproc REST API.

Image versioning: Dataproc's image versioning feature lets you switch between different versions of Apache Spark, Apache Hadoop, and other tools.

Built-in integration: The built-in integration with Cloud Storage, BigQuery, and Cloud Bigtable ensures that data will not be lost. This, together with Cloud Logging and Cloud Monitoring, provides a complete data platform and not just a Spark or Hadoop cluster.

Cluster web interfaces:

Some of the core open source components included with Dataproc clusters, such as Apache Hadoop and Apache Spark, provide web interfaces. These interfaces can be used

to manage and monitor cluster resources and facilities, such as the YARN resource manager, the Hadoop Distributed File System (HDFS), MapReduce, and Spark.

Connecting to web interfaces:

You can connect to web interfaces running on a Dataproc cluster using the Dataproc Component Gateway, your project's Cloud Shell, or the Google Cloud CLI `gcloud` command-line tool.

Dataproc components:

When you create a cluster, standard Apache Hadoop ecosystem components are automatically installed on the cluster. You can install additional components, called "optional components" on the cluster when you create the cluster. Adding optional components to a cluster has the following advantages:

- ✓ Faster cluster startup times
- ✓ Tested compatibility with specific Dataproc versions
- ✓ Use of a cluster parameter instead of an initialization action script
- ✓ Optional components are integrated with other Dataproc components.

Available optional components:

Optional component	COMPONENT_NAME in <code>gcloud</code> commands and API requests
Anaconda	ANACONDA
Docker	DOCKER
Druid	DRUID
Flink	FLINK
HBase	HBASE
Hive WebHCat	HIVE_WEBHCAT
Jupyter Notebook	JUPYTER

Presto	PRESTO
Ranger	RANGER
Solr	SOLR
Zeppelin Notebook	ZEPPELIN
Zookeeper	ZOOKEEPER

Adding optional components:

To create a Dataproc cluster and install one or more optional components on the cluster, use the `gcloud beta dataproc clusters create cluster-name` command with the `--optional-components` flag.

```
gcloud dataproc clusters create cluster-name \
  --optional-components=COMPONENT-NAME(s) \
  ... other flags
```

Dataproc compute options:

Dataproc clusters are built on Compute Engine instances. Machine types define the virtualized hardware resources available to an instance. Compute Engine offers both predefined machine types and custom machine types. Dataproc clusters can use both predefined and custom types for both master and/or worker nodes.

Dataproc supports the following Compute Engine predefined machine types in clusters:

- ✓ General purpose machine types, which include N1, N2, N2D, and E2 machine types:
- ✓ Dataproc also supports N1, N2, N2D, and E2 custom machine types.
- ✓ Compute-optimized machine types, which include C2 machine types.
- ✓ Memory-optimized machine types, which include M1 and M2 machine types.

Custom machine types:

Custom machine types are ideal for the following workloads:

- ✓ Workloads that are not a good fit for the predefined machine types.
- ✓ Workloads that require more processing power or more memory, but don't need all of the upgrades that are provided by the next machine type level.

Dataproc Data Storage:

Dataproc integrates with Apache Hadoop and the Hadoop Distributed File System (HDFS). The following features and considerations can be important when selecting compute and data storage options for Dataproc clusters and jobs:

- ✓ HDFS with Cloud Storage: Dataproc uses the Hadoop Distributed File System (HDFS) for storage. Additionally, Dataproc automatically installs the HDFS-compatible Cloud Storage connector, which enables the use of Cloud Storage in parallel with HDFS. Data can be moved in and out of a cluster through upload/download to HDFS or Cloud Storage.
- ✓ VM disks:
 - By default, when no local SSDs are provided, HDFS data and intermediate shuffle data is stored on VM boot disks, which are Persistent Disks.
 - If you use local SSDs, HDFS data and intermediate shuffle data is stored on the SSDs.
 - Persistent Disk size and type affect performance and VM size, whether using HDFS or Cloud Storage for data storage.
 - VM Boot disks are deleted when the cluster is deleted.

Cloud Dataproc use cases:

1. Log processing
2. Ad-hoc data analysis
3. Machine learning

Dataproc Workflow Templates:

The Dataproc WorkflowTemplates API provides a flexible and easy-to-use mechanism for managing and executing workflows. A Workflow Template is a reusable workflow configuration. It defines a graph of jobs with information on where to run those jobs.

Key Points:

Instantiating a Workflow Template launches a Workflow. A Workflow is an operation that runs a Directed Acyclic Graph (DAG) of jobs on a cluster.

If the workflow uses a managed cluster, it creates the cluster, runs the jobs, and then deletes the cluster when the jobs are finished.

If the workflow uses a cluster selector, it runs jobs on a selected existing cluster.

Workflows are ideal for complex job flows. You can create job dependencies so that a job starts only after its dependencies complete successfully.

When you create a workflow template Dataproc does not create a cluster or submit jobs to a cluster. Dataproc creates or selects a cluster and runs workflow jobs on the cluster when a workflow template is instantiated.

Kinds of Workflow Templates:

Managed cluster:

A workflow template can specify a managed cluster. The workflow will create an "ephemeral" cluster to run workflow jobs, and then delete the cluster when the workflow is finished.

Cluster selector:

A workflow template can specify an existing cluster on which to run workflow jobs by specifying one or more user labels previously attached to the cluster. The workflow will

run on a cluster that matches all of the labels. If multiple clusters match all labels, Dataproc selects the cluster with the most YARN available memory to run all workflow jobs. At the end of workflow, Dataproc does not delete the selected cluster.

Parameterized:

If you will run a workflow template multiple times with different values, use parameters to avoid editing the workflow template for each run:

- ✓ define parameters in the template, then
- ✓ pass different values for the parameters for each run.

Inline:

Workflows can be instantiated inline using the gcloud command with workflow template YAML files or by calling the Dataproc InstantiateInline API. Inline workflows do not create or modify workflow template resources.

BUILD EXTRACT, TRANSFORM, AND LOAD PIPELINES USING CLOUD DATAFLOW

DATAFLOW:

Dataflow is a serverless, fast and cost-effective service that supports both stream and batch processing. It provides portability with processing jobs written using the open source Apache Beam libraries and removes operational overhead from your data engineering teams by automating the infrastructure provisioning and cluster management.

The Apache Beam SDK is an open source programming model that enables you to develop both batch and streaming pipelines. Pipelines are created with an Apache Beam program and then run them on the Dataflow service.

Apache Beam Programming Model:

Apache Beam is an open source, unified model for defining both batch- and streaming-data parallel-processing pipelines. The Apache Beam programming model simplifies the mechanics of large-scale data processing. Using one of the Apache Beam SDKs, you build a program that defines the pipeline. Then, one of Apache Beam's supported distributed processing backends, such as Dataflow, executes the pipeline.

Basic concepts:

Pipelines

A pipeline encapsulates the entire series of computations involved in reading input data, transforming that data, and writing output data. The input source and output sink can be the same or of different types, allowing you to convert data from one format to another. Apache Beam programs start by constructing a Pipeline object, and then using that object as the basis for creating the pipeline's datasets. Each pipeline represents a single, repeatable job.

PCollection

A PCollection represents a potentially distributed, multi-element dataset that acts as the pipeline's data. Apache Beam transforms use PCollection objects as inputs and outputs for each step in your pipeline. A PCollection can hold a dataset of a fixed size or an unbounded dataset from a continuously updating data source.

Transforms

A transform represents a processing operation that transforms data. A transform takes one or more PCollections as input, performs an operation that you specify on each element in that collection, and produces one or more PCollections as output. A transform can perform nearly any kind of processing operation, including performing mathematical computations on data, converting data from one format to another, grouping data together, reading and writing data, filtering

data to output only the elements you want, or combining data elements into single values.

ParDo

ParDo is the core parallel processing operation in the Apache Beam SDKs, invoking a user-specified function on each of the elements of the input PCollection. ParDo collects the zero or more output elements into an output PCollection. The ParDo transform processes elements independently and possibly in parallel.

Pipeline I/O

Apache Beam I/O connectors let you read data into your pipeline and write output data from your pipeline. An I/O connector consists of a source and a sink. All Apache Beam sources and sinks are transforms that let your pipeline work with data from several different data storage formats. You can also write a custom I/O connector.

Aggregation

Aggregation is the process of computing some value from multiple input elements. The primary computational pattern for aggregation in Apache Beam is to group all elements with a common key and window. Then, it combines each group of elements using an associative and commutative operation.

User-defined functions (UDFs)

Some operations within Apache Beam allow executing user-defined code as a way of configuring the transform. For ParDo, user-defined code specifies the operation to apply to every element, and for Combine, it specifies how values should be combined. A pipeline might contain UDFs written in a different language than the language of your runner. A pipeline might also contain UDFs written in multiple languages.

Runner

Runners are the software that accepts a pipeline and executes it. Most runners are translators or adapters to massively parallel big-data processing systems. Other runners exist for local testing and debugging.

Source

A transform that reads from an external storage system. A pipeline typically reads input data from a source. The source has a type, which may be different from the sink type, so you can change the format of data as it moves through the pipeline.

Sink

A transform that writes to an external data storage system, like a file or a database.

Streaming pipelines:

Unbounded PCollections, or unbounded collections, represent data in streaming pipelines. An unbounded collection contains data from a continuously updating data source such as Pub/Sub.

To aggregate elements in unbounded collections, windows, watermarks, and triggers can be used.

The concept of windows also applies to bounded PCollections that represent data in batch pipelines.

Windows and windowing functions:

Windowing functions divide unbounded collections into logical components, or windows. Windowing functions group unbounded collections by the timestamps of the individual elements. Each window contains a finite number of elements.

You set the following windows with the Apache Beam SDK or Dataflow SQL streaming extensions:

- Tumbling windows (called fixed windows in Apache Beam)
- Hopping windows (called sliding windows in Apache Beam)
- Session windows

Watermarks:

A watermark is a threshold that indicates when Dataflow expects all of the data in a window to have arrived. If new data arrives with a timestamp that's in the window but older than the watermark, the data is considered late data.

Dataflow tracks watermarks because of the following:

- Data is not guaranteed to arrive in time order or at predictable intervals.
- Data events are not guaranteed to appear in pipelines in the same order that they were generated.

The data source determines the watermark. You can allow late data with the Apache Beam SDK. Dataflow SQL does not process late data.

Triggers:

Triggers determine when to emit aggregated results as data arrives. By default, results are emitted when the watermark passes the end of the window.

You can use the Apache Beam SDK to create or modify triggers for each collection in a streaming pipeline. You cannot set triggers with Dataflow SQL.

The Apache Beam SDK can set triggers that operate on any combination of the following conditions:

- Event time, as indicated by the timestamp on each data element.
- Processing time, which is the time that the data element is processed at any given stage in the pipeline.
- The number of data elements in a collection.

Dataflow Templates:

Dataflow templates allow you to package a Dataflow pipeline for deployment. Anyone with the correct permissions can then use the template to deploy the packaged pipeline. You can create your own custom Dataflow templates, and Google provides pre-built templates for common scenarios.

Advantages of pipeline templates

Templates have several advantages over directly deploying a pipeline to Dataflow:

- Templates separate pipeline design from deployment. For example, a developer can create a template, and a data scientist can deploy the template at a later time.
- Templates can have parameters that let you customize the pipeline when you deploy the template.
- You can deploy a template by using the Google Cloud console, the Google Cloud CLI, or REST API calls. You don't need a development environment or any pipeline dependencies installed on your local machine.
- A template is a code artifact that can be stored in a source control repository and used in continuous integration (CI/CD) pipelines.

Dataflow supports two types of template: Flex templates, which are newer, and classic templates. If you are creating a new Dataflow template, we recommend creating it as a Flex template.

Template workflow

Using Dataflow templates involves the following high-level steps:

1. Developers set up a development environment and develop their pipeline. The environment includes the Apache Beam SDK and other dependencies.
2. Depending on the template type (Flex or classic):
 - For Flex templates, the developers package the pipeline into a Docker image, push the image to Container Registry or Artifact Registry, and upload a template specification file to Cloud Storage.
 - For classic templates, developers run the pipeline, create a template file, and stage the template to Cloud Storage.
3. Other users submit a request to the Dataflow service to run the template.
4. Dataflow creates a pipeline from the template.

BIGQUERY:

- ✓ BigQuery is a fully managed enterprise data warehouse that helps you manage and analyze your data with built-in features like machine learning, geospatial analysis, and business intelligence.
- ✓ BigQuery's serverless architecture allows to use SQL queries to answer your organization's biggest questions with zero infrastructure management.
- ✓ BigQuery maximizes flexibility by separating the compute engine that analyzes your data from your storage choices.
- ✓ BigQuery interfaces include Google Cloud console interface and the BigQuery command-line tool.
- ✓ Developers and data scientists can use client libraries with familiar programming including Python, Java, JavaScript, and Go, as well as BigQuery's REST API and RPC API to transform and manage data.

BigQuery storage:

- ✓ BigQuery stores data using a columnar storage format that is optimized for analytical queries.
- ✓ BigQuery presents data in tables, rows, and columns and provides full support for database transaction semantics (ACID).
- ✓ BigQuery storage is automatically replicated across multiple locations to provide high availability.

BigQuery administration:

- BigQuery provides centralized management of data and compute resources while Identity and Access Management (IAM) helps to secure those resources with the access model that's used throughout Google Cloud.

Loading data into BigQuery:

There are several ways to ingest data into BigQuery:

- Batch load a set of data records.
- Stream individual records or batches of records.
- Use queries to generate new data and append or overwrite the results to a table.
- Use a third-party application or service.

Batch loading:

With batch loading, you load the source data into a BigQuery table in a single batch operation.

Options for batch loading in BigQuery include the following:

- **Load jobs.** Load data from Cloud Storage or from a local file by creating a load job.

- **SQL.** The LOAD DATA SQL statement loads data from one or more files into a new or existing table.
- **BigQuery Data Transfer Service.** Use [BigQuery Data Transfer Service](#) to automate loading data from Google Software as a Service (SaaS) apps or from third-party applications and services.
- **BigQuery Storage Write API.** The [Storage Write API](#) lets you batch-process an arbitrarily large number of records and commit them in a single atomic operation. If the commit operation fails, you can safely retry the operation. Unlike BigQuery load jobs, the Storage Write API does not require staging the data to intermediate storage such as Cloud Storage.
- **Other managed services.** Use other managed services to export data from an external data store and import it into BigQuery.

Batch loading can be done as a one-time operation or on a recurring schedule.

Streaming:

With streaming, you continually send smaller batches of data in real time, so the data is available for querying as it arrives. Options for streaming in BigQuery include the following:

- **Storage Write API.** The Storage Write API supports high-throughput streaming ingestion with exactly-once delivery semantics.
- **Dataflow.** Use Dataflow with the Apache Beam SDK to set up a streaming pipeline that writes to BigQuery.
- **BigQuery Connector for SAP.** The BigQuery Connector for SAP enables near real time replication of SAP data directly into BigQuery.

Generated data:

You can use SQL to generate data and store the results in BigQuery. Options for generating data include:

- Use data manipulation language (DML) statements to perform bulk inserts into an existing table or store query results in a new table.
- Use a CREATE TABLE ... AS statement to create a new table from a query result.
- Run a query and save the results to a table. You can append the results to an existing table or write to a new table.

BigQuery analytics:

BigQuery is optimized to run analytic queries on large datasets, including terabytes of data in seconds and petabytes in minutes. Understanding its capabilities and how it processes queries can help you maximize your data analysis investments.

Analytic workflows:

BigQuery supports several data analysis workflows:

Ad hoc analysis. BigQuery uses Google Standard SQL, the SQL dialect in BigQuery, to support ad hoc analysis.

Geospatial analysis. BigQuery uses geography data types and Google Standard SQL geography functions to let you analyze and visualize geospatial data.

Machine learning. BigQuery ML uses Google Standard SQL queries to let you create and execute machine learning (ML) models in BigQuery.

Business intelligence. BigQuery BI Engine is a fast, in-memory analysis service that lets you build rich, interactive dashboards and reports without compromising performance, scalability, security, or data freshness.

Queries:

The primary unit of analysis in BigQuery is the SQL query. BigQuery has two SQL dialects: Google Standard SQL and legacy SQL. Google Standard SQL is the preferred dialect.

- **Data sources:**

BigQuery lets you query the following types of data sources:

Data stored in BigQuery - load data into BigQuery for analysis.

External data - query various external data sources such as other Google Cloud storage services (like Cloud Storage) or database services (like Cloud Spanner or Cloud SQL).

Multi-cloud data - query data that's stored in other public clouds such as AWS or Azure.

Public datasets - can analyze any of the datasets that are available in the public dataset marketplace.

- **Types of queries:**

After you load your data into BigQuery, you can query the data using one of the following query job types:

- **Interactive query jobs.** By default, BigQuery runs interactive (on-demand) query jobs as soon as possible.
- **Batch query jobs.** With these jobs, BigQuery queues each batch query on your behalf and then starts the query when idle resources are available, usually within a few minutes.

INTRODUCTION TO MACHINE LEARNING IN THE CLOUD

Machine learning allows businesses to enable the data to teach the system how to solve the problem at hand with machine learning algorithms and how to get better over time.

Machine learning (ML) is a subfield of artificial intelligence (AI). The goal of ML is to make computers learn from the data that you give them. Instead of writing code that describes the action the computer should take, your code provides an algorithm that adapts based on examples of intended behavior. The resulting program, consisting of the algorithm and associated learned parameters, is called a trained model.

- Artificial intelligence, or AI, is an umbrella term that includes anything related to computers mimicking human intelligence. For example, in an online word processor, robots perform human actions for spelling and grammar checks.
- Machine learning is a toolset, like Newton's laws of mechanics. Just as you can use Newton's laws to learn how long it will take a ball to fall to the ground if you drop it off a cliff, you can use machine learning to solve certain kinds of problems at scale by using data examples, but without the need for custom code.
- You might have also heard the term deep learning, or deep neural networks. Deep learning is a subset of machine learning that adds layers in between input data and output results to make a machine learn at more depth. It's a type of machine learning that works even when the data is unstructured, like images, speech, video, natural language text, and so on. Image classification is a type of deep learning. A machine can learn how to classify images into categories when it is shown lots of examples.

The basic difference between machine learning and other techniques in AI is that in machine learning, machines learn. They don't start out intelligent, they become intelligent.

So, how do machines become intelligent? Intelligence requires training. To train a machine learning model, examples are required.

For example, to train a model to estimate how much you'll owe in taxes, you must show the model many, many examples of tax returns or if you want to train a model to estimate trip time between one location and another, you'll need to show it many examples of previous journeys.

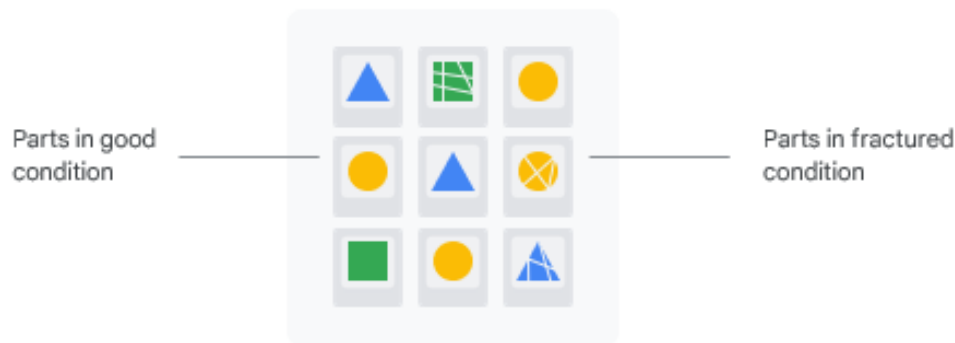
The first stage of ML is to train an ML model with examples.

Imagine you work for a manufacturing company and you want to train a machine learning model to detect defects in the parts before they are assembled into products. You'd start by creating a dataset of images of parts. Some of those parts would be good and some parts would be defective. For each image, you will assign a corresponding label and use that set of examples to train the model.



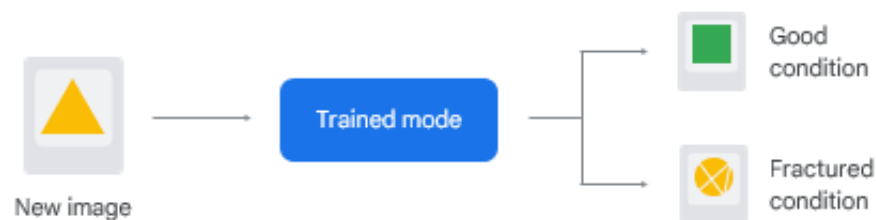
An important detail to emphasize is that a machine learning model is only as good as the data used to train it. And a good model requires a lot of training data of historical examples of rejected parts and parts in good condition. With these elements, you can train a model to categorize parts as defective or not.

The basic reason why ML models need high-quality data is because they don't have human general knowledge; data is the only thing they have access to.



After the model has been trained, it can be used to make predictions on data it has never seen before.

In this example, the input for the trained model is an image of a part. Because the model has been trained on specific examples of good and defective parts, it can correctly predict that this part is in good condition.



Standard algorithms use cases

Algorithms, or ML models, are standard. That means that they exist independently of the use case. Although detecting manufacturing defects in images and detecting diseased leaves in images are two different use cases, the same algorithm, which is an image classification network, works for both.

Similarly, standard algorithms predict the future value of a time series or to transcribe human speech or text. ResNet, for example, is a standard algorithm for image classification. It's not essential to understand how an image classification algorithm works, only that it's the algorithm that we should use if we want to classify images of automotive parts.



ML OPTIONS ON GOOGLE CLOUD

Google Cloud offers four options for building machine learning models.

1. The first option is **BigQuery ML**. You'll remember from the previous module of this course that BigQuery ML is a tool for using SQL queries to create and execute machine learning models in BigQuery. If you already have your data in BigQuery and your problems fit the predefined ML models, this could be your best choice.
2. The second option is **AutoML**, which is a no-code solution, so you can build your own machine learning models on Vertex AI through a point-and-click interface.
3. The third option is **custom training**, through which you can code your own machine learning environment, the training, and the deployment, which provides you with flexibility and control over the ML pipeline.
4. And finally, there are **pre-built APIs**, which are application programming interfaces. This option lets you use machine learning models that have already

been built and trained by Google, so you don't have to build your own machine learning models if you don't have enough training data or sufficient machine learning expertise in-house.

Big Query ML

BigQuery ML lets you create and execute machine learning models in BigQuery using standard SQL queries. BigQuery ML democratizes machine learning by letting SQL practitioners build models using existing SQL tools and skills. BigQuery ML increases development speed by eliminating the need to move data.

BigQuery ML functionality is available by using:

- The Google Cloud console
- The bq command-line tool
- The BigQuery REST API
- An external tool such as a Jupyter notebook or business intelligence platform

Machine learning on large datasets requires extensive programming and knowledge of ML frameworks. These requirements restrict solution development to a very small set of people within each company, and they exclude data analysts who understand the data but have limited machine learning knowledge and programming expertise.

BigQuery ML empowers data analysts to use machine learning through existing SQL tools and skills. Analysts can use BigQuery ML to build and evaluate ML models in BigQuery. Analysts don't need to export small amounts of data to spreadsheets or other applications or wait for limited resources from a data science team.

Supported models in BigQuery ML

A model in BigQuery ML represents what an ML system has learned from the training data.

BigQuery ML supports the following types of models:

Linear regression for forecasting; for example, the sales of an item on a given day. Labels are real-valued (they cannot be +/- infinity or NaN).

Binary logistic regression for classification; for example, determining whether a customer will make a purchase. Labels must only have two possible values.

Multiclass logistic regression for classification. These models can be used to predict multiple possible values such as whether an input is "low-value," "medium-value," or "high-value." Labels can have up to 50 unique values. In BigQuery ML, multiclass logistic regression training uses a multinomial classifier with a cross-entropy loss function.

K-means clustering for data segmentation; for example, identifying customer segments. K-means is an unsupervised learning technique, so model training does not require labels nor split data for training or evaluation.

Matrix Factorization for creating product recommendation systems. You can create product recommendations using historical customer behavior, transactions, and product ratings and then use those recommendations for personalized customer experiences.

Time series for performing time-series forecasts. You can use this feature to create millions of time series models and use them for forecasting. The model automatically handles anomalies, seasonality, and holidays.

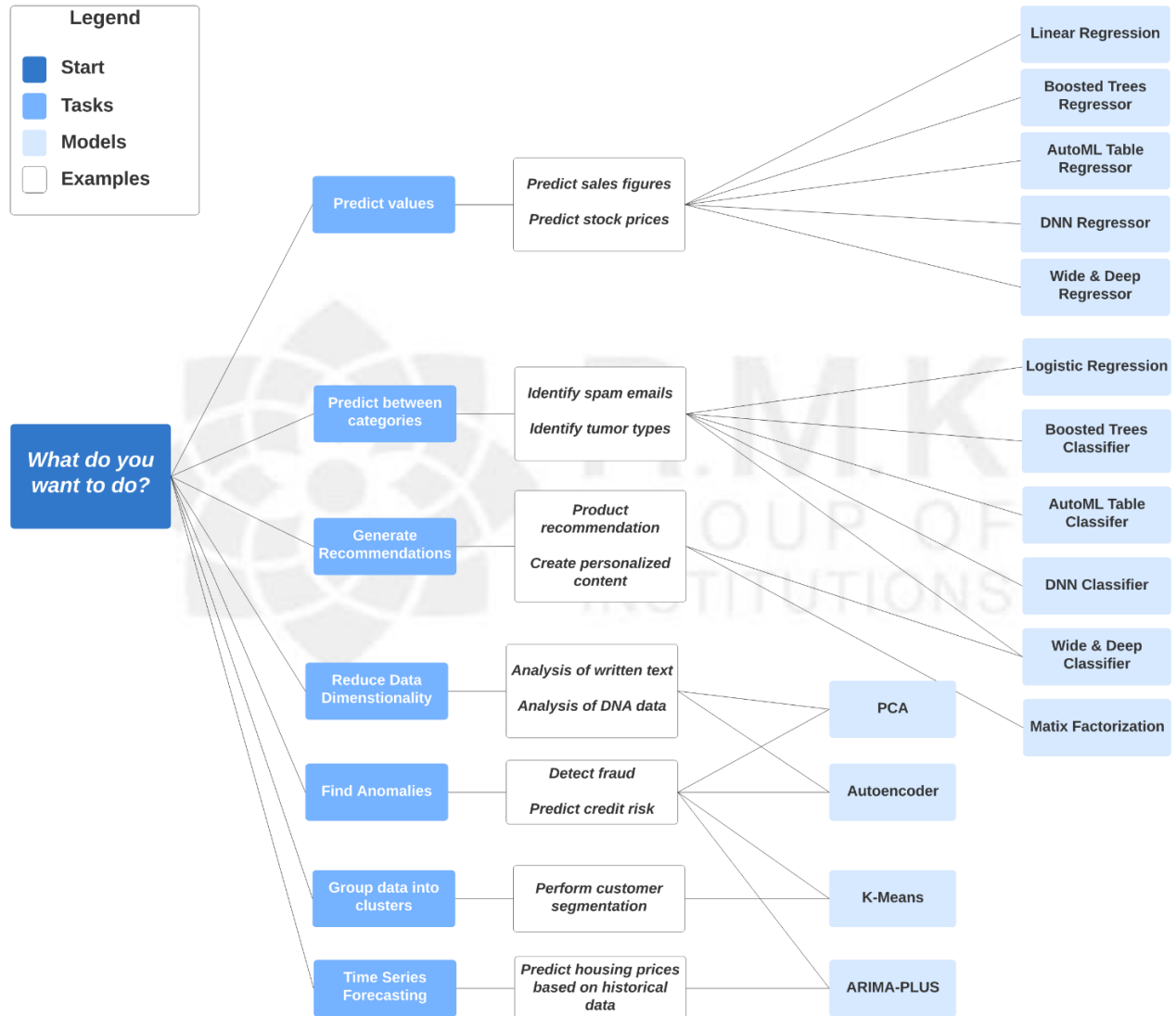
Boosted Tree for creating XGBoost based classification and regression models.

Deep Neural Network (DNN) for creating TensorFlow-based Deep Neural Networks for classification and regression models.

AutoML Tables to create best-in-class models without feature engineering or model selection. AutoML Tables searches through a variety of model architectures to decide the best model.

TensorFlow model importing. This feature lets you create BigQuery ML models from previously trained TensorFlow models, then perform prediction in BigQuery ML.

Autoencoder for creating Tensorflow-based BigQuery ML models with the support of sparse data representations. The models can be used in BigQuery ML for tasks such as unsupervised anomaly detection and non-linear dimensionality reduction.



Model Selection Guide

Advantages of BigQuery ML

BigQuery ML has the following advantages over other approaches to using ML with a cloud-based data warehouse:

1. BigQuery ML democratizes the use of ML by empowering data analysts, the primary data warehouse users, to build and run models using existing business intelligence tools and spreadsheets. Predictive analytics can guide business decision-making across the organization.
2. There is no need to program an ML solution using Python or Java. Models are trained and accessed in BigQuery using SQL—a language data analysts know.
3. BigQuery ML increases the speed of model development and innovation by removing the need to export data from the data warehouse. Instead, BigQuery ML brings ML to the data. The need to export and reformat data has the following disadvantages:
 - Increases complexity because multiple tools are required.
 - Reduces speed because moving and formatting large amounts data for Python-based ML frameworks takes longer than model training in BigQuery.
 - Requires multiple steps to export data from the warehouse, restricting the ability to experiment on your data.
 - Can be prevented by legal restrictions such as HIPAA guidelines.

Steps to create Machine Learning models in BigQuery:

1. Create your dataset
2. Create your model
3. Get training statistics
4. Evaluate your model
5. Use your model to predict outcomes

VERTEX AI

Build, deploy, and scale machine learning (ML) models faster, with fully managed ML tools for any use case. Vertex AI allows users to build machine learning models with either AutoML, a no-code solution, or Custom Training, a code-based solution. AutoML is easy to use and lets data scientists spend more time turning business problems into ML solutions, but custom training enables data scientists to have full control over the development environment and process.

Vertex AI brings AutoML and AI Platform together into a unified API, client library, and user interface. AutoML lets you train models on image, tabular, text, and video datasets without writing code, while training in AI Platform lets you run custom training code. With Vertex AI, both AutoML training and custom training are available options. Whichever option you choose for training, you can save models, deploy models, and request predictions with Vertex AI.

Google's solution to many of the production and ease-of-use challenges is Vertex AI, a unified platform that brings all the components of the machine learning ecosystem and workflow together.

Vertex AI is a unified platform

Vertex AI is a unified platform, it means having one digital experience to create, manage, and deploy models over time, and at scale. In Vertex AI Machine Learning process carried out by 4 stages

- **Data readiness:**

During the data readiness stage, users can upload data from wherever it's stored: Cloud Storage, BigQuery, or a local machine.

- **Feature readiness:**

Then, **during the feature readiness stage**, users can create features, which are the processed data that will be put into the model, and then share them with others by using the feature store.

- **Training and hyperparameter tuning:**

After that, it's time for **training and hyperparameter tuning**. This means that when the data is ready, users can experiment with different models and adjust hyperparameters.

- **Deployment and model monitoring:**

And finally, during **deployment and model monitoring**, users can set up the **pipeline** to transform the model into production by automatically monitoring and performing continuous improvements.

Benefits of Vertex AI

- It's seamless. Vertex AI provides a smooth user experience from uploading and preparing data all the way to model training and production.
- It's scalable. The machine learning operations (MLOps) provided by Vertex AI helps to monitor and manage the ML production and therefore scale the storage and computing power automatically.
- It's sustainable. All the artifacts and features created with Vertex AI can be reused and shared.
- And it's speedy. Vertex AI produces models that have 80% fewer lines of code than competitors

Components of Vertex AI

This section describes the pieces that make up Vertex AI and the primary purpose of each piece.

- **Model training**

You can train models on Vertex AI by using AutoML, or if you need the wider range of customization options available in AI Platform Training, use custom training.

In custom training, you can select from among many different machine types to power your training jobs, enable distributed training, use hyperparameter tuning, and accelerate with GPUs.

- **Model deployment for prediction**

You can deploy models on Vertex AI and get an endpoint to serve predictions on Vertex AI whether or not the model was trained on Vertex AI.

- **Vertex AI Data Labeling**

Data Labeling jobs let you request human labeling for a dataset that you plan to use to train a custom machine learning model. You can submit a request to label your video, image, or text data.

To submit a labeling request, you provide a representative sample of labeled data, specify all the possible labels for your dataset, and provide some instructions for how to apply those labels. The human labelers follow your instructions, and when the labeling request is complete, you get your annotated dataset that you can use to train a machine learning model.

- **Vertex AI Feature Store**

Vertex AI Feature Store is a fully managed repository where you can ingest, serve, and share ML feature values within your organization. Vertex AI Feature Store manages all the underlying infrastructure. For example, it provides storage and compute resources for you and can easily scale as needed.

Tools to interact with Vertex AI

- **The Google Cloud console**

You can deploy models to the cloud and manage your datasets, models, endpoints, and jobs on the Google Cloud console. This option provides a user interface for working with your machine learning resources. As part of Google Cloud, your Vertex AI resources are connected to useful tools like Cloud Logging and Cloud Monitoring. The best place to start using the Google Cloud console is the Dashboard page of the Vertex AI section

- **Cloud Client Libraries**

Vertex AI provides client libraries for some languages to help you make calls to the Vertex AI API. The client libraries provide an optimized developer experience by using each supported language's natural conventions and styles.

Alternatively, you can use the Google API Client Libraries to access the Vertex AI API by using other languages, such as Dart. When using the Google API Client Libraries, you build representations of the resources and objects used by the API. This is easier and requires less code than working directly with HTTP requests.

- **REST API**

The Vertex AI REST API provides RESTful services for managing jobs, models, and endpoints, and for making predictions with hosted models on Google Cloud.

- **Vertex AI Workbench notebooks instances**

Vertex AI Workbench is a Jupyter notebook-based development environment for the entire data science workflow. You can interact with Vertex AI services from within a Vertex AI Workbench instance's Jupyter notebook. Vertex AI Workbench integrations and features can make it easier to access your data, process data faster, schedule notebook runs, and more.

- **Deep Learning VM Images**

Deep Learning VM Images is a set of virtual machine images optimized for data science and machine learning tasks. All images come with key ML frameworks and

tools pre-installed. You can use them out of the box on instances with GPUs to accelerate your data processing tasks.

Deep Learning VM images are available to support many combinations of framework and processor. There are currently images supporting TensorFlow Enterprise, TensorFlow, PyTorch, and generic high-performance computing, with versions for both CPU-only and GPU-enabled workflows.

- **Deep Learning Containers**

Deep Learning Containers are a set of Docker containers with key data science frameworks, libraries, and tools pre-installed. These containers provide you with performance-optimized, consistent environments that can help you prototype and implement workflows quickly.

Vertex AI workflow

Vertex AI uses a standard machine learning workflow:

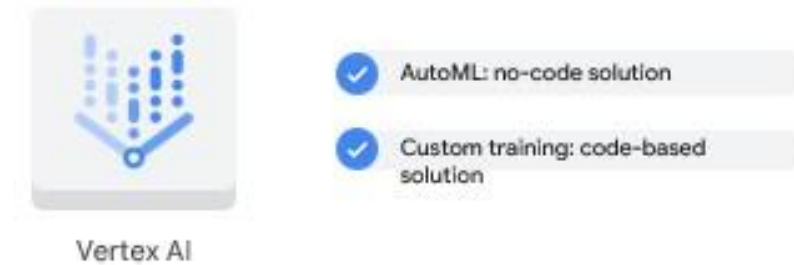
- Gather your data: Determine the data you need for training and testing your model based on the outcome you want to achieve.
- Prepare your data: Make sure your data is properly formatted and labeled.
- Train: Set parameters and build your model.
- Evaluate: Review model metrics.
- Deploy and predict: Make your model available to use.

But before you start gathering your data, you need to think about the problem you are trying to solve. This will inform your data requirements.

Vertex AI provides two solutions in one

Vertex AI allows users to build machine learning models with either **AutoML**, a no-code solution, or **Custom Training**, a code-based solution. AutoML is easy to use and lets data scientists spend more time turning business problems into ML solutions, but custom

training enables data scientists to have full control over the development environment and process.



AUTO ML

AutoML is a no-code solution to build ML models through Vertex AI. It allows users to train high-quality custom machine learning models with minimal effort or machine learning expertise.

If you've worked with ML models before, you know that training and deploying ML models can be incredibly time consuming, because you need to repeatedly add new data and features, try different models, and tune parameters to achieve the best result.

To solve this problem, when AutoML was first announced in January of 2018, the goal was to automate machine learning pipelines to save data scientists from manual work, such as tuning hyperparameters and comparing against multiple models.

Two technologies Used:

- **Transfer Learning:**

In transfer learning, you build a knowledge base in the field. You can think of this like gathering many books to create a library.

Transfer learning is a powerful technique that lets people with smaller datasets, or less computational power, achieve state-of-the-art results by using pre-trained models that

have been trained on similar, larger datasets. Because the model learns through transfer learning, it doesn't have to learn from the beginning, so it can generally reach higher accuracy with much less data and computation time than models that don't use transfer learning.

- **Neural architect search:**

The second technology is **neural architect search**. The goal of neural architect search is to find the optimal model for the relevant project.

AutoML is powered by the latest machine-learning research, so although a model performs training, the AutoML platform actually trains and evaluates multiple models and compares them to each other. This neural architecture search produces an ensemble of ML models and chooses the best one.

AutoML supports four types of data: **image**, **tabular**, **text**, and **video**. For each data type, AutoML solves different types of problems, called **objectives**.

Benefits of AutoML

- ✓ Using these technologies has produced a tool that can significantly benefit data scientists
- ✓ One of the biggest benefits is that it's a **no-code solution**. That means it can train high-quality custom machine learning models with minimal effort and requires little machine learning expertise.
- ✓ This allows data scientists to focus their time on tasks like defining business problems or evaluating and improving model results.
- ✓ Others might find AutoML useful to quickly prototype models and explore new datasets before investing in development. This might involve using it to identify the best features in a dataset

Difference between Traditional ML Development and Auto ML

Auto ML, which is short for automated machine learning.

In traditional ML models, training and deploying ML can be incredibly time consuming, because you need to repeatedly add new data and features, try different models, and tune parameters to achieve the best result.

To solve this problem, when AutoML was first announced in January of 2018, the goal was to automate machine learning pipelines to save data scientists from manual work, such as tuning hyperparameters and comparing against multiple models.



The AutoML product offerings are:

- Sight — AutoML Vision, AutoML Video Intelligence
- Language — AutoML Natural Language, AutoML Translation
- Structured data — AutoML Tables

How AutoML works?

Here's the components you need:

Google Account + Enable Billing

Cloud AutoML and Storage API

AutoML Tables takes your dataset and starts training for multiple model architectures at the same time. This lets cloud AutoML determine the best model architecture for your

data quickly, without having to serially iterate over the many possible model architectures. The model architectures to test include:

- Linear
- Feedforward deep neural network
- Gradient Boosted Decision Tree
- AdaNet
- Ensembles of various model architectures

As new model architectures come out of the research community, Google will add those as well.



Dataset → AutoML → Generate Predictions with a REST API call. Image Source, Cloud AutoML

Data Preparation

Before we can start training we must ensure that the training data is in a format that is supported by the platform. AutoML Tabular supports two ways of importing training data; if we are already utilising Google's BigQuery platform, we can import the data straight

from a BigQuery table, or we can upload a comma-separated values (CSV) file to Cloud Storage.

CSV format	- The delimiter must be a comma (",") - The first line of the first file must be a header. - Column names can include any alphanumeric character or an underscore. The column name cannot begin with an underscore.
Dataset size	- At most 100 GB of data, chunked in files of at most 10 GB per file. - There must be at least 1,000 and no more than 100,000,000 rows. - At least 2 and no more than 1,000 columns, with one being the target variable.
Target variable	- Must be categorical or numerical. - If it is categorical, there must be at least 2 and no more than 500 categories.

If you are dealing with structured data, AutoML Tables is recommended. Depending on use cases, AutoML Tables will create the necessary model to solve the problem:

- A binary classification model predicts a binary outcome (one of two classes). Use this for yes or no questions, for example, predicting whether a customer would buy a subscription (or not). All else being equal, a binary classification problem requires less data than other model types.
- A multi-class classification model predicts one class from three or more discrete classes. Use this to categorize things. For the retail example, you'd want to build a multi-class classification model to segment customers into different personas.
- A regression model predicts a continuous value. For the retail example, you'd also want to build a regression model to forecast customer spending over the next month.

Creating a Dataset

Importing a dataset can be done through the Google Cloud Platform (GCP) Console. In the Vertex AI section there is a Datasets page where we can manage our datasets. From there we can create a tabular dataset and select a data source for the dataset.



Item ID	Date and Time	Brand	Category	Price
COLD43245	2019-03-31 11:54:12	Brrr Brands	Coat	\$259.57
ACCS54326	2019-04-01 14:12:10	Top Tops	Hat	\$49.99
BOOT12365	2019-04-01 14:31:34	Head Over Heels	Shoe	\$89.49
RING54903	2019-04-02 18:01:59	Stone Gold Jewelry	Jewelry	\$189.51

Sample structured data that you can use with Cloud AutoML



A visualisation of the splits, Image Source: About Train, Validation and Test Sets in Machine Learning

On the other hand, working with unstructured data is more complicated. The Vision API classifies images into thousands of predefined categories, detects individual objects and faces within images, and finds and reads printed words contained within images. If you want to detect individual objects, faces, and text in your dataset, or your need for image classification is quite generic, try out the Vision API and see if it works for you.



The Vision API classification images

Coupled with image labeling in the Vision API, AutoML Vision enables you to perform supervised learning, which involves training a computer to recognize patterns from labeled data. Using supervised learning, we can train a model to recognize the patterns and content that we care about in images.

Training a Model

Once the dataset has been created we can start training our first model.

# Rows	Suggested Training Time
< 100.000	1-3 hours
100.000 - 1.000.000	1-6 hours
1.000.000 - 10.000.000	1-12 hours
> 10.000.000	3-24 hours

Vision API

			
Property 88%	Landmark 94%	House 94%	Property 92%
House 87%	Sky 93%	Home 92%	Building 84%
Architecture 85%	House 89%	Property 92%	Town 84%
Home 81%	Building 89%	Real Estate 80%	House 83%

AutoML

			
Tudor .5	Tudor .1	Tudor .2	Tudor 92
Neoclassical .1	Victorian .94	Neoclassical .3	Neoclassical .4
Modern .89	Modern .3	Modern .3	Modern .2
Ranch .5	Ranch .1	Ranch .93	Ranch .2

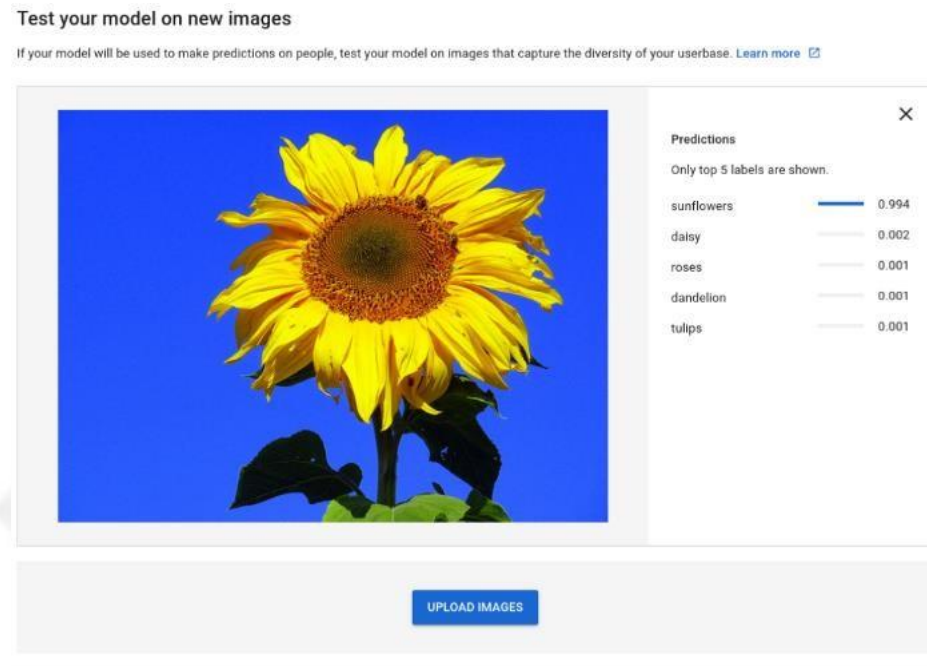
The model outputs a series of numbers that communicate how strongly it associates each label with that example. If the number is high, the model has high confidence that the label should be applied to that document.

After you have created (trained) a model, you can request a prediction for an image using the predict method. The available options for online (individual) prediction are:

- Web UI
- Command-Line
- Python
- Java
- Node.js

Deployment, Testing and Explainability

The trained model that is produced by AutoML can then be deployed in two ways; we can export the model as a saved TensorFlow (TF) model which we can then serve ourselves in a Docker container, or we can deploy the model directly to a GCP endpoint.



CUSTOM TRAINING

Custom training allows the ML environment to control the entire ML development process by developing customers own code, starting from data preparation to model deployment.

If you want to code your machine learning model, you can use this option by building a custom training solution with Vertex AI Workbench.

Workbench is a **single development environment** for the entire data science workflow, from exploring, to training, and then deploying a machine learning model with code.

Before any coding begins, customer must determine what environment want for their ML training code to use.

Choose a training code structure

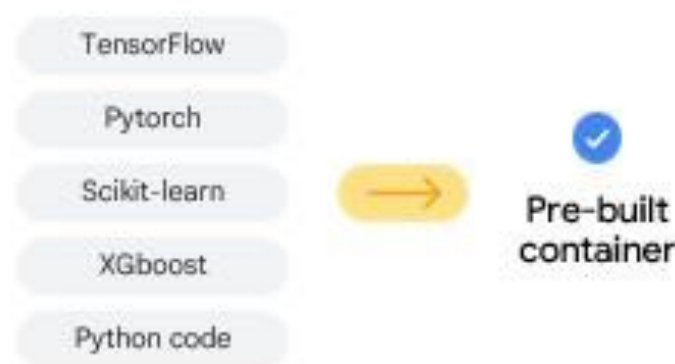
First, determine what structure you want your ML training code to take. You can provide training code to Vertex AI in one of the following forms:

- **Pre-built container**
- **Custom container**

Pre-built container

Imagine that a container is a kitchen. A pre-built container would represent a fully furnished room with cabinets and appliances (which represent the dependencies) that includes all the cookware (which represents the libraries) you need to make a meal.

So, if your ML training needs a platform like TensorFlow, Pytorch, scikit-learn, or XGboost, and Python code to work with the platform, a pre-built container is probably your best solution



A Python training application to use with a pre-built container.

Create a Python source distribution with code that trains an ML model and exports it to Cloud Storage. This training application can use any of the dependencies included in the pre-built container that you plan to use it with.

Custom container

A custom container, alternatively, is like an empty room with no cabinets, appliances, or cookware.

Create a Docker container image with code that trains an ML model and exports it to Cloud Storage. Include any dependencies required by your code in the container image.

Use this option if you want to use dependencies that are not included in one of the Vertex AI pre-built containers for training. For example, if you want to train using a Python ML framework that is not available in a pre-built container, or if you want to train using a programming language other than Python, then this is the better option.

Pre-built APIs

When using AutoML, you define a domain-specific labeled training dataset that is used to create the custom ML model you require. However, if you don't need a domain-specific dataset, Google's suite of pre-built ML APIs might meet your needs.

In this section, you'll explore some of those APIs.

Good machine learning models require lots of high-quality training data. You should aim for hundreds of thousands of records to train a custom model. If you don't have that kind of data, pre-built APIs are a great place to start.

Pre-built APIs are offered as services. Often they can act as building blocks to create the application you want without the expense or complexity of creating your own models. They save the time and effort of building, curating, and training a new dataset, so you can directly deal with predictions.

Examples of Pre-built API

- The **Speech-to-Text API** converts audio to text for data processing.
- The **Cloud Natural Language API** recognizes parts of speech called entities and sentiment.
- The **Cloud Translation API** converts text from one language to another.
- The **Text-to-Speech API** converts text into high-quality voice audio.
- The **Vision API** works with and recognizes content in static images.
- And the **Video Intelligence API** recognizes motion and action in video.

And Google has already done a lot of work to train these models by using Google datasets. For example

- The Vision API is based on Google's image datasets.
- The Speech-to-Text API is trained on YouTube captions.
- And the Translation API is built on Google Translate.

You recall that how well a model is trained depends on how much data is available to train it. As you might expect, Google has many images, text, and ML researchers to train its pre-built models. This means less work for you.

1.Speech-to-Text API

Speech-to-Text enables easy integration of Google speech recognition technologies into developer applications. Send audio and receive a text transcription from the Speech-to-Text API service.

Speech requests

Speech-to-Text has three main methods to perform speech recognition. These are listed below:

Synchronous Recognition (REST and gRPC) sends audio data to the Speech-to-Text API, performs recognition on that data, and returns results after all audio has been processed. Synchronous recognition requests are limited to audio data of 1 minute or less in duration.

Asynchronous Recognition (REST and gRPC) sends audio data to the Speech-to-Text API and initiates a Long Running Operation. Using this operation, you can periodically poll for recognition results. Use asynchronous requests for audio data of any duration up to 480 minutes.

Streaming Recognition (gRPC only) performs recognition on audio data provided within a gRPC bi-directional stream. Streaming requests are designed for real-time recognition purposes, such as capturing live audio from a microphone. Streaming recognition provides interim results while audio is being captured, allowing result to appear, for example, while a user is still speaking.

Requests contain configuration parameters as well as audio data. The following sections describe these type of recognition requests, the responses they generate, and how to handle those responses in more detail.

Speech-to-Text API recognition

A Speech-to-Text API synchronous recognition request is the simplest method for performing recognition on speech audio data. Speech-to-Text can process up to 1 minute of speech audio data sent in a synchronous request. After Speech-to-Text processes and recognizes all of the audio, it returns a response.

A synchronous request is blocking, meaning that Speech-to-Text must return a response before processing the next request. Speech-to-Text typically processes audio

faster than realtime, processing 30 seconds of audio in 15 seconds on average. In cases of poor audio quality, your recognition request can take significantly longer.

2. Cloud Natural Language API

Provides natural language understanding technologies, such as sentiment analysis, entity recognition, entity sentiment analysis, and other text annotations, to developers.

Service: language.googleapis.com

To call this service, we recommend that you use the Google-provided client libraries. If your application needs to use your own libraries to call this service, use the following information when you make the API requests.

Discovery document

A Discovery Document is a machine-readable specification for describing and consuming REST APIs. It is used to build client libraries, IDE plugins, and other tools that interact with Google APIs. One service may provide multiple discovery documents. This service provides the following discovery documents

[https://language.googleapis.com/\\$discovery/rest?version=v1](https://language.googleapis.com/$discovery/rest?version=v1)

[https://language.googleapis.com/\\$discovery/rest?version=v1beta2](https://language.googleapis.com/$discovery/rest?version=v1beta2)

Service endpoint

A service endpoint is a base URL that specifies the network address of an API service. One service might have multiple service endpoints. This service has the following service endpoint and all URIs below are relative to this service endpoint
<https://language.googleapis.com>.

3. Text-to-Speech API

Text-to-Speech converts text or Speech Synthesis Markup Language (SSML) input into audio data of natural human speech like MP3 or LINEAR16 (the encoding used in WAV files).

Text-to-Speech allows developers to create natural-sounding, synthetic human speech as playable audio. You can use the audio data files you create using Text-to-Speech to power your applications or augment media like videos or audio recordings.

Text-to-Speech is ideal for any application that plays audio of human speech to users. It allows you to convert arbitrary strings, words, and sentences into the sound of a person speaking the same things.

Imagine that you have a voice assistant app that provides natural language feedback to your users as playable audio files. Your app might take an action and then provide human speech as feedback to the user.

Speech synthesis

The process of translating text input into audio data is called synthesis and the output of synthesis is called synthetic speech.

Text-to-Speech takes two types of input:

- Raw text
- SSML-formatted data (discussed below).

To create a new audio file, you call the synthesise endpoint of the API.

The speech synthesis process generates raw audio data as a base64-encoded string. You must decode the base64-encoded string into an audio file before an application can play it. Most platforms and operating systems have tools for decoding base64 text into playable media files.

Voices

Text-to-Speech creates raw audio data of natural, human speech. That is, it creates audio that sounds like a person talking. When you send a synthesis request to Text-to-Speech, you must specify a voice that 'speaks' the words.

Text-to-Speech has a wide selection of custom voices available for you to use. The voices differ by language, gender, and accent (for some languages). For example, you can create audio that mimics the sound of a female English speaker with a British accent. You can also convert the same text into a different voice, say a male English speaker with an Australian accent.

4.Cloud Translation API

Cloud Translation enables your websites and applications to dynamically translate text programmatically through an API. Translation uses a Google pre-trained or a custom machine learning model to translate text. By default, Translation uses a Google pre-trained Neural Machine Translation (NMT) model, which Google updates on semi-regular cadence when more training data or better techniques become available.

Benefits

Cloud Translation can translate text for more than 100 language pairs. If you don't know the language of your source text, Cloud Translation can detect it for you.

Cloud Translation scales seamlessly and allows unlimited character translations per day. However, there are restrictions on content size for each request and request rates. Additionally, you can use quota limits to manage your budget.

Cloud Translation editions

Cloud Translation offers two editions:

Cloud Translation - Basic (v2)

Cloud Translation - Advanced (v3)

Both editions support language detection and translation. They also use the same Google pre-trained model to translate content. Cloud Translation - Advanced includes features such as batch requests, AutoML custom models, and glossaries.

Cloud Translation - Basic (v2)

Cloud Translation - Basic enables you to dynamically translate between languages by using Google's pre-trained Neural Machine Translation (NMT) model. Basic is optimized for simplicity and scale. It's a good fit for applications that handle primarily casual user-generated content, such as chat, social media, or comments.

Cloud Translation - Advanced (v3)

Cloud Translation - Advanced has all of the capabilities of Cloud Translation - Basic and more. Cloud Translation - Advanced edition is optimized for customization and long form content use cases. Customization features include glossaries and custom model selection.

Pricing

Translation charges you on a monthly basis based on the number of characters that you send.

5. Vision API

Cloud Vision allows developers to easily integrate vision detection features within applications, including image labeling, face and landmark detection, optical character recognition (OCR), and tagging of explicit content.

Vision API currently allows you to use the following features:

- **Text detection**

Optical character recognition (OCR) for an image; text recognition and conversion to machine-coded text. Identifies and extracts UTF-8 text in an image.

- **Document text detection (dense text / handwriting)**

Optical character recognition (OCR) for a file (PDF/TIFF) or dense text image;
dense text recognition and conversion to machine-coded text.

- **Landmark detection**

Landmark Detection detects popular natural and human-made structures within an image.

- **Logo Detection**

Logo Detection detects popular product logos within an image.

- **Label detection**

Provides generalized labels for an image.

For each label returns a textual description, confidence score, and topicality rating.

- **Image properties**

Returns dominant colors in an image.

Each color is represented in the RGBA color space, has a confidence score, and displays the fraction of pixels occupied by the color [0, 1].

- **Object localization**

The Vision API can detect and extract multiple objects in an image with Object Localization.

- **Crop hint detection**

Provides a bounding polygon for the cropped image, a confidence score, and an importance fraction of this salient region with respect to the original image for each request.

You can provide up to 16 image ratio values (width:height) for a single image.

- **Web entities and pages**

Provides a series of related Web content to an image.

- **Face detection**

Locates faces with bounding polygons, and identifies specific facial "landmarks" such as eyes, ears, nose, mouth, etc. along with their corresponding confidence values.

Returns likelihood ratings for emotion (joy, sorrow, anger, surprise) and general image properties (underexposed, blurred, headwear present).

Likelihoods ratings are expressed as 6 different values: UNKNOWN, VERY_UNLIKELY, UNLIKELY, POSSIBLE, LIKELY, or VERY_LIKELY.

6.Video Intelligence API

The Video Intelligence API allows developers to use Google video analysis technology as part of their applications. The REST API enables users to annotate videos stored locally or in Cloud Storage, or live-streamed, with contextual information at the level of the entire video, per segment, per shot, and per frame.



10. ASSIGNMENT

1. **Implement a Spark-based big data processing job using Google Dataproc or AWS EMR and describe your approach.**
2. **Build a basic ETL pipeline using Google Dataflow or AWS Glue to process large-scale sales data and store it in a data warehouse.**
3. **A logistics company needs real-time tracking of its delivery fleet. Suggest a cloud-based streaming solution using Google Pub/Sub**
4. **. What is data pipeline orchestration, and how do tools like Google Cloud Composer help in automating workflows?**
5. **Discuss real-world use cases of big data managed services in industries like healthcare, finance, and retail.**

11. PART A QUESTIONS AND ANSWERS

1. What is Big Data?

Big Data is a collection of datasets that is large and complex, making it very difficult to process using legacy data processing applications. Tools for big data can help with the volume of the data collected, the speed at which that data becomes available to an organization for analysis, and the complexity or varieties of that data.

2. What are the characteristics of Big Data?

The attributes that define big data are volume, variety, velocity, and variability. These big data attributes are commonly referred to as the four v's.

Volume: The key characteristic of big data is its scale—the volume of data that is available for collection by your enterprise from a variety of devices and sources.

Variety: Variety refers to the formats that data comes in, such as email messages, audio files, videos, sensor data, and more. Classifications of big data variety include structured, semi-structured, and unstructured data.

Velocity: Big data velocity refers to the speed at which large datasets are acquired, processed, and accessed.

Variability: Big data variability means the meaning of the data constantly changes. Therefore, before big data can be analyzed, the context and meaning of the datasets must be properly understood.

3. What is Cloud Dataproc?

Cloud Dataproc is a managed Apache Hadoop and Spark service for batch processing, querying, streaming and machine learning. Users can quickly spin up Hadoop or Spark clusters and resize them at any time without compromising data pipelines through automation and orchestration. It can be fully integrated with other Google big data services, such as BigQuery and Bigtable, as well as Stackdriver Logging and Monitoring.

4. What is Cloud Dataflow?

Cloud Dataflow is a serverless stream and batch processing service. Users can build a pipeline to manage and analyze data in the cloud, while Cloud Dataflow automatically manages the resources. It was built to integrate with other Google services, including BigQuery and Cloud Machine Learning, as well as third-party products, such as Apache Spark and Apache Beam.

5. What is Google BigQuery?

BigQuery is a data warehouse that processes and analyzes large data sets using SQL queries. These services can capture and examine streaming data for real-time analytics. It stores data with Google's Capacitor columnar data format, and users can load data via streaming or batch loads. To load, export, query and copy data, use the classic web UI, the web UI in the GCP Console, the bq command-line tool or client libraries. Since BigQuery is a serverless offering, enterprises only pay for the storage and compute they consume.

6. What is Apache Hadoop?

Apache Hadoop is a set of tools and technologies which enables a cluster of computers to store and process large volumes of data. The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models.

7. List the features of Cloud Dataproc.

- Cost Effective
- Fast and Scalable
- Open source ecosystem
- Fully managed
- Image versioning
- Built-in integration

8. How will you create a Dataproc cluster with an optional component?

To create a Dataproc cluster and install one or more optional components on the cluster, use the `gcloud beta dataproc clusters create cluster-name` command with the `-optional-components` flag.

```
gcloud dataproc clusters create cluster-name \
  --optional-components=COMPONENT-NAME(s) \
  ... other flags
```

9. List the usecases of Dataproc.

- Log processing
- Ad-hoc data analysis
- Machine learning

10. What is Dataproc workflow template?

The Dataproc WorkflowTemplates API provides a flexible and easy-to-use mechanism for managing and executing workflows. A Workflow Template is a reusable workflow configuration. It defines a graph of jobs with information on where to run those jobs.

11. List the types of workflow templates.

- Managed cluster
- Cluster selector
- Parameterized
- Inline

12. What is a pipeline?

A pipeline encapsulates the entire series of computations involved in reading input data, transforming that data, and writing output data. The input source and output sink can be the same or of different types, allowing you to convert data from one format to another. Apache Beam programs start by constructing a Pipeline object, and then using that object as the basis for creating the pipeline's datasets. Each pipeline represents a single, repeatable job.

13. What is BigQuery storage?

- ✓ BigQuery stores data using a columnar storage format that is optimized for analytical queries.
- ✓ BigQuery presents data in tables, rows, and columns and provides full support for database transaction semantics (ACID).
- ✓ BigQuery storage is automatically replicated across multiple locations to provide high availability.

14. What are the ways to ingest data into BigQuery?

There are several ways to ingest data into BigQuery:

- Batch load a set of data records.
- Stream individual records or batches of records.
- Use queries to generate new data and append or overwrite the results to a table.
- Use a third-party application or service.

15. Define query. What are the types of data sources?

The primary unit of analysis in BigQuery is the SQL query. BigQuery has two SQL dialects: Google Standard SQL and legacy SQL. Google Standard SQL is the preferred dialect.

▪ Data sources:

BigQuery lets you query the following types of data sources:

Data stored in BigQuery - load data into BigQuery for analysis.

External data - query various external data sources such other Google Cloud storage services (like Cloud Storage) or database services (like Cloud Spanner or Cloud SQL).

Multi-cloud data - query data that's stored in other public clouds such as AWS or Azure.

Public datasets - can analyze any of the datasets that are available in the public dataset marketplace.

16. Define Machine Learning

Machine learning (ML) is a subfield of artificial intelligence (AI). The goal of ML is to make computers learn from the data that you give them. Instead of writing code that describes the action the computer should take, your code provides an algorithm that adapts based on examples of intended behavior. The resulting program, consisting of the algorithm and associated learned parameters, is called a trained model.

17. List out the ML options available in Google Cloud

- BigQuery ML
- Auto ML
- Custom training
- Pre-built APIs

18. Describe Auto ML option in Google Cloud

AutoML is a no-code solution to build ML models through Vertex AI. It allows users to train high-quality custom machine learning models with minimal effort or machine learning expertise.

19. What are the Ways BigQuery ML functionality can accessed by customer?

- The Google Cloud console
- The bq command-line tool
- The BigQuery REST API
- An external tool such as a Jupyter notebook or business intelligence platform

20. Define VERTEX AI.

Build, deploy, and scale machine learning (ML) models faster, with fully managed ML tools for any use case. Vertex AI allows users to build machine learning models with either AutoML, a no-code solution, or Custom Training, a code-based solution. AutoML is easy to use and lets data scientists spend more time turning business problems into ML solutions, but custom training enables data scientists to have full control over the development environment and process.

21. List out and describe the stages in VERTEX AI.

- Data readiness:

During the data readiness stage, users can upload data from wherever it's stored: Cloud Storage, BigQuery, or a local machine.

- Feature readiness:

Then, during the feature readiness stage, users can create features, which are the processed data that will be put into the model, and then share them with others by using the feature store.

- Training and hyperparameter tuning:

After that, it's time for training and hyperparameter tuning. This means that when the data is ready, users can experiment with different models and adjust hyperparameters.

- Deployment and model monitoring:

And finally, during deployment and model monitoring, users can set up the pipeline to transform the model into production by automatically monitoring and performing continuous improvements.

22. What are the Benefits of Vertex AI

- It's seamless. Vertex AI provides a smooth user experience from uploading and preparing data all the way to model training and production.
- It's scalable. The machine learning operations (MLOps) provided by Vertex AI helps to monitor and manage the ML production and therefore scale the storage and computing power automatically.
- It's sustainable. All the artifacts and features created with Vertex AI can be reused and shared.

- And it's speedy. Vertex AI produces models that have 80% fewer lines of code than competitors

23. Which of the architecture the Auto ML will support

- Linear
- Feedforward deep neural network
- Gradient Boosted Decision Tree
- AdaNet
- Ensembles of various model architectures

24. Benefits of AutoML

- ✓ Using these technologies has produced a tool that can significantly benefit data scientists
- ✓ One of the biggest benefits is that it's a no-code solution. That means it can train high-quality custom machine learning models with minimal effort and requires little machine learning expertise.
- ✓ This allows data scientists to focus their time on tasks like defining business problems or evaluating and improving model results.
- ✓ Others might find AutoML useful to quickly prototype models and explore new datasets before investing in development. This might involve using it to identify the best features in a dataset

25. Examples of Pre-built API:

- The Speech-to-Text API converts audio to text for data processing.
- The Cloud Natural Language API recognizes parts of speech called entities and sentiment.
- The Cloud Translation API converts text from one language to another.
- The Text-to-Speech API converts text into high-quality voice audio.
- The Vision API works with and recognizes content in static images.
- And the Video Intelligence API recognizes motion and action in video.

12. PART B QUESTIONS

1. How will you leverage Big Data operations with cloud Dataproc?
2. How will you build extract, transform, and load pipelines using cloud dataflow?
3. Explain in detail about BigQuery.
4. Explain in detail about Machine Learning tools in Google Cloud.
5. Demonstrate the concept VERTEX AI in detail.
6. List out the Prebuilt APIs and explain it in detail.
7. Explain the concept Auto ML in detail.



13. ONLINE CERTIFICATIONS

1. Cloud Digital Leader

[Cloud Digital Leader](#) | [Google Cloud](#)

2. Associate Cloud Engineer:

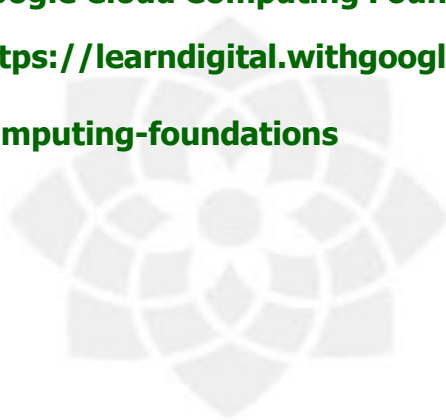
[Associate Cloud Engineer Certification](#) | [Google Cloud](#)

3. Google Cloud Computing Foundations Course

https://onlinecourses.nptel.ac.in/noc20_cs55/preview

4. Google Cloud Computing Foundations

<https://learndigital.withgoogle.com/digitalgarage/course/gcloud-computing-foundations>



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14. REAL TIME APPLICATIONS

1. Real-Time Fraud Detection in Banking & Finance

How it Works: Google Cloud processes real-time transactions to detect fraud patterns.

Google Cloud Services Used:

BigQuery ML → Runs ML models on large datasets

Cloud Pub/Sub → Ingests real-time transaction data

Cloud Dataflow → Processes data streams for anomalies

Example: A bank uses BigQuery ML to analyze millions of daily transactions and detect fraudulent activities in real-time.

2. Personalized Content Recommendation (Streaming & E-commerce)

How it Works: Google Cloud analyzes user preferences to recommend products or content.

Google Cloud Services Used:

Recommendations AI → ML-based product recommendations

BigQuery → Stores and queries user behavior data

Vertex AI → Trains ML models for personalization

Example: YouTube and Google Play Store use Google Cloud AI models to recommend videos, apps, and music based on user history.

3. Predictive Maintenance in Manufacturing & IoT

How it Works: IoT sensors collect machine data, and Google Cloud predicts failures before they happen.

Google Cloud Services Used:

Cloud IoT Core → Collects sensor data in real-time

BigQuery → Analyzes massive IoT datasets

Vertex AI → Trains models to predict machine failures

Example: A car manufacturing company predicts equipment failures in assembly lines to avoid costly downtime.

4. Smart Healthcare & Disease Prediction

How it Works: Google Cloud analyzes electronic health records (EHRs) and medical images for diagnosis.

Google Cloud Services Used:

Healthcare API → Securely stores and processes medical data

AutoML Vision → Analyzes medical images (e.g., X-rays)

BigQuery ML → Predicts disease trends from patient data

Example: Hospitals use Google Cloud AI to detect cancer in X-ray images with high accuracy.

5. Real-Time Traffic Management & Navigation

How it Works: Google Cloud processes GPS and weather data to provide real-time traffic insights.

Google Cloud Services Used:

BigQuery GIS → Analyzes geospatial traffic data

Cloud Pub/Sub → Streams real-time GPS data

Cloud Dataflow → Processes and updates traffic conditions

Example: Google Maps and Waze use Google Cloud Big Data services to provide real-time route optimization.

15. ASSESSMENT SCHEDULE

Tentative schedule for the Assessment During 2023-2024 Even semester

S.NO	Name of the Assessment	Start Date	End Date	Portion
1	IAT 1	27.01.2025	01.02.2025	UNIT 1 & 2
2	IAT 2	10.03.2025	15.03.2025	UNIT 3 & 4
3	REVISION	-	-	UNIT 5 , 1 & 2
4	MODEL	03.04.2025	17.04.2025	ALL 5 UNITS

17. PRESCRIBED TEXT BOOKS AND REFERENCES

REFERENCES:

1. <https://cloud.google.com/docs>
2. https://www.cloudskillsboost.google/course_templates/153
3. <https://nptel.ac.in/courses/106105223>



18. MINI PROJECT

1. Real-Time Sentiment Analysis on Social Media

Objective: Analyze tweets, Facebook comments, or YouTube reviews to determine public sentiment.

Tools: Google Cloud Natural Language API, BigQuery, Pub/Sub, Dataflow

Key Features:

- Collect real-time data from Twitter API
- Perform sentiment analysis using ML
- Store results in BigQuery and visualize in Google Data Studio

2. Predictive Maintenance in Manufacturing (IoT + ML)

Objective: Use sensor data from industrial machines to predict failures before they happen

Tools: Google Cloud IoT Core, BigQuery ML, Vertex AI

Key Features:

- Collect sensor data from IoT devices
- Use ML models to predict equipment failures
- Reduce downtime with predictive analytics

3. Movie Recommendation System using BigQuery ML

Objective: Build a personalized movie recommendation engine like Netflix.

Tools: Google BigQuery ML, AutoML Tables, Cloud Storage

Key Features:

- Train a collaborative filtering model using user preferences
- Store and process large-scale movie data
- Provide real-time recommendations

4. Credit Card Fraud Detection

Objective: Detect fraudulent transactions using real-time data analysis.

Tools: Google BigQuery ML, Cloud Pub/Sub, Dataflow

Key Features:

Train ML models on transaction history

Detect anomalies in real-time payments

Send alerts when fraud is suspected

5. Healthcare Disease Prediction Using ML

Objective: Predict diseases like diabetes or heart disease from patient health records.

Tools: Google Cloud Healthcare API, Vertex AI, AutoML Tables

Key Features:

Use electronic health records (EHRs) for prediction

Train ML models to detect disease risks

Provide early warnings for patients

Thank you



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